Trend research — 2026-04-19 · incr 19

12-contributor degeneracy audit: 2 silent-fail, sharp directional-accuracy hierarchy from 58% to 98%

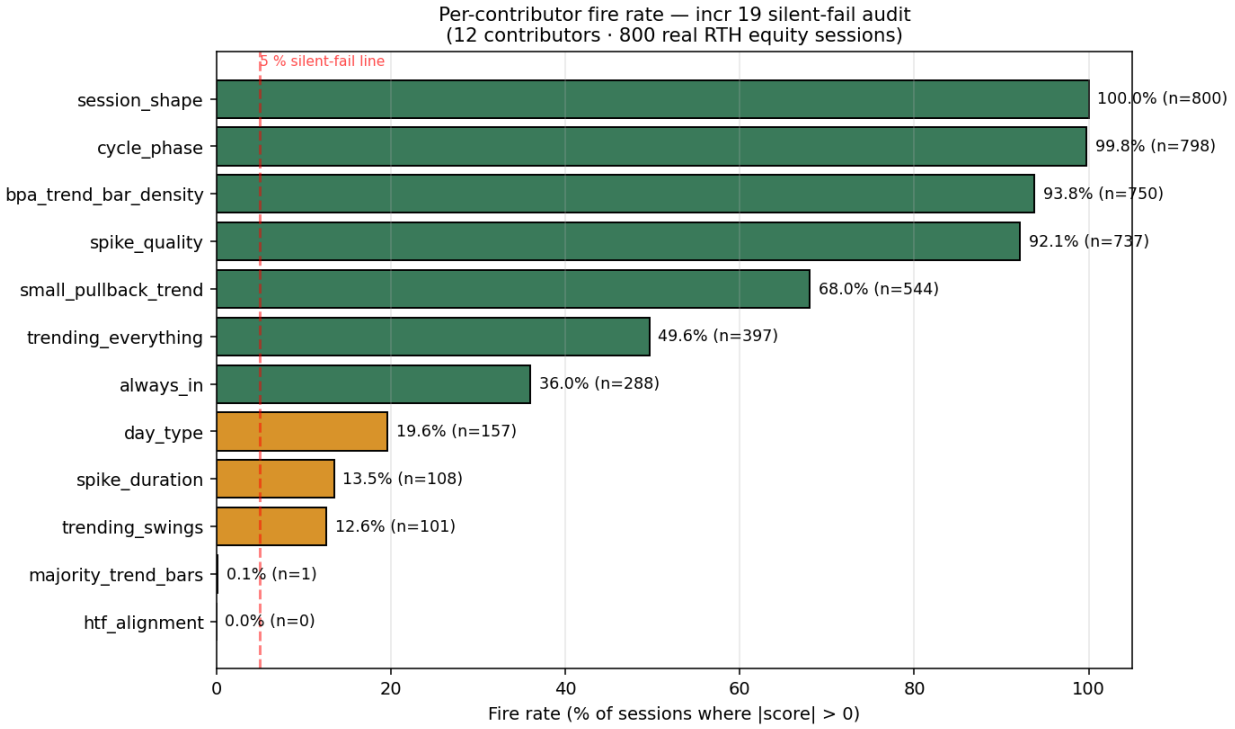
TL;DR

Incr 17 caught **majority_trend_bars** as constant 0 on real data. This run asks: are *other* contributors silently failing the same way? Ran a full degeneracy audit — fire rate, unique-value count, output entropy, directional accuracy — across all 12 contributors on the same 800-session real RTH equity sample. **Silent-fail (< 5% fires): majority_trend_bars**, **htf_alignment**. **Very low entropy (< 0.5 bits): majority_trend_bars**, **htf_alignment**. Read-only. No production code changed.

Per-contributor diagnostic table

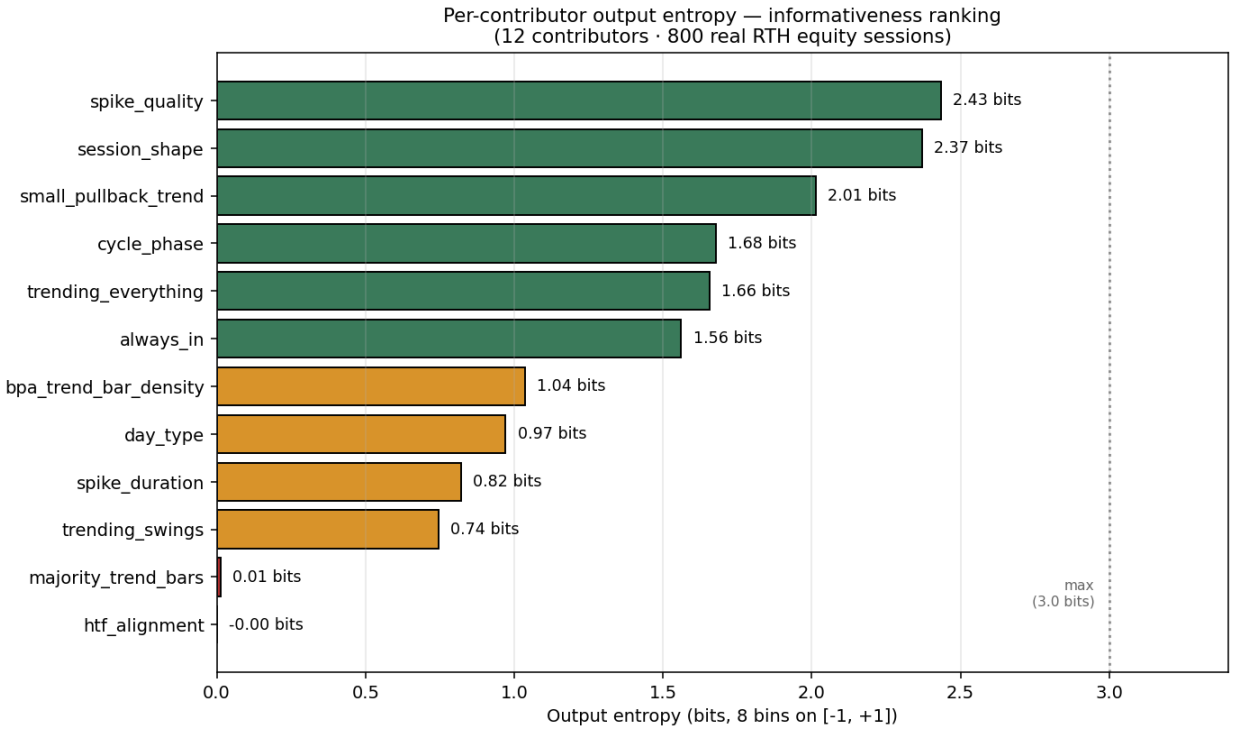
contributor	fire	uniq	ent	sat	dirAcc
htf_alignment	0.0%	1	-0.00	0%	n/a
majority_trend_bars	0.1%	2	0.01	0%	0.0%
trending_swings	12.6%	5	0.74	14%	75.0%
spike_duration	13.5%	5	0.82	71%	85.7%
day_type	19.6%	84	0.97	4%	75.2%
always_in	36.0%	8	1.56	0%	57.6%
trending_everything	49.6%	5	1.66	89%	97.9%
small_pullback_trend	68.0%	123	2.01	0%	68.8%
spike_quality	92.1%	32	2.43	3%	64.1%
bpa_trend_bar_density	93.8%	117	1.04	0%	81.2%
cycle_phase	99.8%	443	1.68	0%	65.6%
session_shape	100.0%	644	2.37	0%	96.3%

Fire rate ranking



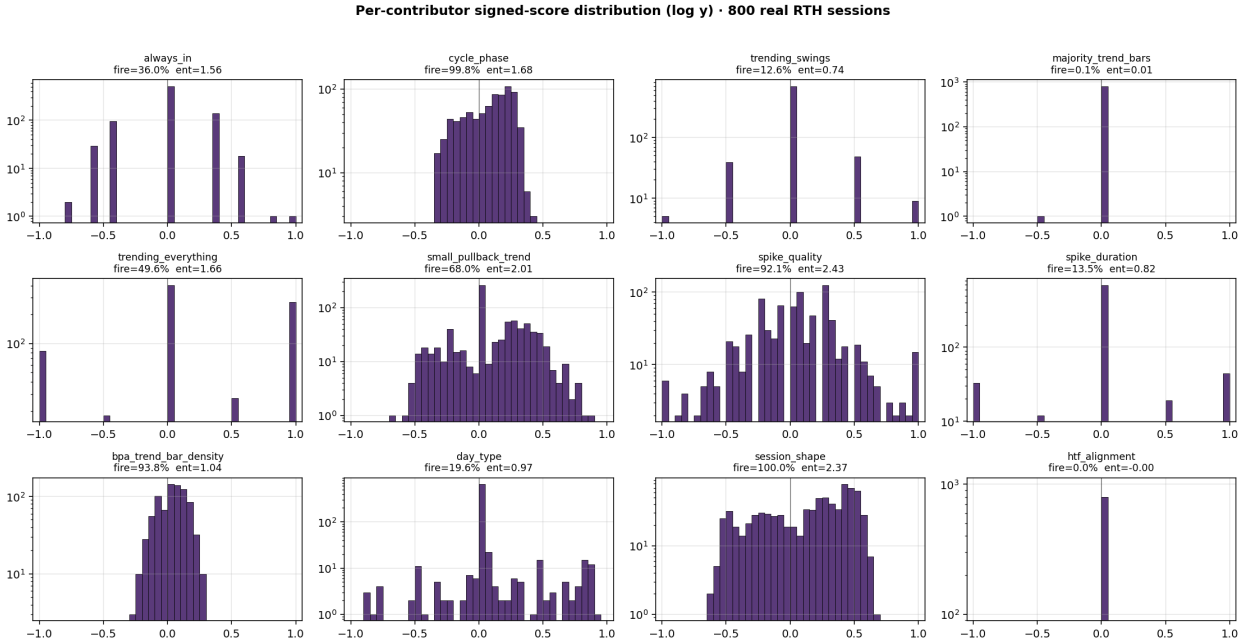
Each bar = one contributor's fire rate (% of 800 sessions where |signed score| > 0). Red = silent fail (<5%). Orange = sparse (<30%). Green = healthy.

Output entropy ranking



Shannon entropy in bits, 8 equal bins on [-1, +1]. Max = 3.0 bits (uniform). Red = degenerate (<0.5 bits, the contributor mass piles into one bucket).

Per-contributor signed-score distributions



Twelve panels (log y-axis). Each panel = one contributor's per-session score distribution across the 800 RTH equity sessions. Spike-at-zero = silent fail mode.

What's next — needs Will's nod

- Confirms the incr-18 majority_trend_bars finding is the worst case but **likely not the only case**. Each contributor flagged red above is a candidate for the same kind of floor / threshold recalibration study incr 18 ran on majority_trend_bars.
- **Recommended next study:** for each silent-fail contributor, run an incr-18-style two-axis sweep to identify the binding parameter (threshold vs floor vs gate) and quantify the directional accuracy trade-off at lower thresholds.
- Pattern Lab DB backfill still blocking the WR-by-setup-type test.
- Multi-month / overnight ES sample still pending — same caveat as incr 18.
- **Do not flip any contributor threshold without your nod** — per the scanner CLAUDE.md standing direction.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification